

Composite-likelihood demographic inference for the Scottish wildcat and the domestic cat from a folded joint site frequency spectrum

Timothy Wood

University of Bristol

Supervisors: Mark Beaumont, Dennis Prangle, Grace Yan

September 2026

Abstract

Three two-population demographic models were fitted to a folded joint site frequency spectrum constructed from 16 wild-caught Scottish wildcats and 6 domestic cats, using 22.5 Mb of callable sequence on chromosomes A1 and A2 and dadi version 2.4.4. Because dadi maximises a composite likelihood, standard model-selection criteria and confidence intervals are over-optimistic; models were therefore compared using the composite-likelihood AIC (CLAIC), and confidence intervals were obtained from the Godambe information matrix, with both estimated from 100 block-bootstrap replicates. A model in which the two populations were isolated and then came back into contact fitted substantially worse, by at least 806 CLAIC units, than two models in which changes in population size and migration were placed at domestication and at the loss of the British land bridge. Of these two, the model with continuous size change terminated with a parameter on its bound rather than at an interior optimum. Since CLAIC is not defined at a bound, the model with instantaneous size changes is reported as the primary result. This model places the split between the *silvestris* and *lybica* lineages at approximately 485,000 years ago and the size change in the domestic lineage at approximately 10,600 years ago. Four parameters are estimated with narrow confidence intervals; all others are estimated only to within a factor of four or more. The size change in the wildcat lineage is placed at 673 years ago, a date that corresponds to no known event. The correlation matrix indicates that these interval widths should not be interpreted independently: the deeper parameters lie along a single ridge in the likelihood, and the two most precisely estimated parameters are constrained only in combination.

Contents

1	Introduction	4
2	Methods	4
2.1	Data and constants	4
2.2	Population definitions	5
2.3	Spectrum construction	6
2.4	Models	7
2.5	Optimisation	10
2.6	Model comparison and uncertainty	11
2.7	Conversion to physical units	12
3	Results	13
3.1	Model comparison	13
3.2	Optimiser diagnostics	13
3.3	Parameter estimates	13
3.4	Effect of the step size on the Godambe correction	15
3.5	Fit to the spectrum	16
3.6	Parameter correlations	17
4	Discussion	19
4.1	What the model comparison establishes	19
4.2	Can the historical labels be tested?	20
4.3	What the data determines	20
4.4	Introgression into the Scottish population	21
4.5	The growth fit	22
4.6	The baseline for simulation-based inference	23

4.7	Limitations and extensions	23
5	Conclusion	24
A	Code availability	28
B	PCA plots	29
C	Convergence diagnostics	30
D	Correlation matrices for growth and sec_contact	31

1 Introduction

This report fits and compares three two-population demographic models for the Scottish wildcat (*Felis silvestris*) and the domestic cat (*F. catus*), using a folded joint site frequency spectrum (JSFS) constructed from the first two chromosomes of a 46-cat dataset. The report is intended for an internal audience, so background on the biology and on demographic inference is assumed, and the emphasis is on the methods and their interpretation.

The project began as a follow-up to a preliminary analysis of twelve Scottish wildcat and twelve domestic cat haplotypes over a subset of chromosomes A1 and A2, in which an isolation-with-migration (IM) model appeared to fit poorly. Early exploratory fits to the full dataset appeared to favour a model in which the two populations were isolated for a period before coming back into contact, and the IM model was not carried forward. These exploratory fits were made before the optimisation and model-comparison procedures described in Sections 2.5 and 2.6 had been established, and they are not reported here. In hindsight, dropping IM leaves the comparison without its most natural baseline, since all three of the models that were retained impose more structure than IM does. This point is returned to in Section 4.7.

Aims

1. To test whether a model whose parameters correspond to real events in the history of the two species, namely domestication and the loss of the British land bridge, fits better than a model with a single size per branch and a single change in gene flow.
2. To determine which parameters these data can estimate precisely and which they cannot.
3. To produce a composite-likelihood baseline, with its uncertainties fully documented, against which simulation-based inference methods can later be checked.

2 Methods

2.1 Data and constants

The data consisted of 46 diploid cats (92 haplotypes) genotyped on chromosomes A1 and A2 (NC_018723.3 and NC_018724.3), supplied as MSMC multihetsep files. This format was used because it records callability explicitly: the third column of each file gives the number of sites that passed the masks since the previous segregating site, so that monomorphic sites are counted directly rather than assumed. Summing this column across both chromosomes gives the callable length, $L = 22,488,648$, which is needed for the conversion from θ to N_{ref} (Section 2.7). The data contain 145,716 segregating sites, of which 204 were multi-allelic and were removed, leaving 145,512 biallelic sites.

The mutation rate was set to $\mu = 0.86 \times 10^{-8}$ per site per generation (Wang et al., 2022), a pedigree-based estimate made directly in domestic cats rather than a rate borrowed from another species. Generation time was taken to be 3 years, following Howard-McCombe et al. (2021), who in turn take the value from Beaumont et al. (2001) and Nussberger et al. (2018). Both constants act only as multipliers: they set the scale of the results in years and in individuals, but have no effect on the shape of the spectrum or on the model fits themselves.

2.2 Population definitions

The 46 cats fall into four groups (Table 1). Throughout this report the sampled populations are referred to as Scottish wildcats (*F. silvestris*) and domestic cats (*F. catus*), whereas the branches of the models are named for the lineages they represent: *silvestris* for the wildcat branch and *lybica* for the domestic branch, after the African lineage *F. lybica* from which the domestic cat descends (Driscoll et al., 2007). The distinction matters because the early size of the domestic branch, N_L , refers to the lineage ancestral to the domestic cat rather than to the domestic cat itself; the two are separated in the models by the time T_D .

Table 1: Sample composition and haplotype column assignment. Group membership was checked against the sample manifest and the metadata accompanying the SNP dataset, and independently by PCA (Appendix B), which separates the domestic, Scottish and mainland European groups but does not resolve the division within the Scottish sample.

Group	Individuals	Haplotype columns
Scottish wild-caught	16	0–31
Scottish captive	10	32–51
Domestic	6	52–63
European wild	14	64–91
Total	46	0–91

The Scottish sample is not homogeneous. Hudson’s F_{ST} between the wild-caught and captive animals is 0.085, roughly half of the differentiation between the Scottish and mainland European wildcats (0.169). The captive animals also have lower diversity, with nucleotide diversity π 21% below that of the wild-caught animals, and a higher F_{ST} to every other group, which is consistent with a small, closed breeding population that has undergone genetic drift.

This matters because dadi treats each population as a single, randomly mating group, and sampling from a structured population is known to produce a spurious signal of population size change (Chikhi et al., 2010; Mazet et al., 2016). Folded joint spectra were therefore built under both definitions of the Scottish population, pooled and wild-caught only. Both were brought to 32 Scottish haplotypes, so that any difference between them reflects which animals were included rather than the sample size (Table 2).

Table 2: Effect of the definition of the Scottish population on the folded spectrum. The pooled spectrum was built at 52 Scottish haplotypes and projected down to 32, so that the two spectra compared in the lower rows have the same shape, 32 Scottish and 12 domestic haplotypes; the wild-only spectrum is at its full sample size and needs no projection. Site counts after projection are expectations under hypergeometric subsampling and are rounded to the nearest integer. Bin indices refer to the folded one-dimensional Scottish marginal spectrum.

	pooled (0–51)	wild-only (0–31)
Segregating sites at full sample size	124,321 (52 haplotypes)	123,348 (32 haplotypes)
Segregating sites at 32 haplotypes	121,505 (projected)	123,348
Tajima’s D (Scottish)	−0.804	−0.990
F_{ST} (Scottish, domestic)	0.392	0.384
Intermediate-frequency mass (bins 8–16)	0.157	0.125

Adding the 20 captive haplotypes raises the number of segregating sites by only 973, from 123,348 to 124,321, because the captive animals contribute little polymorphism that the wild-caught animals do not already carry. After projection to 32 haplotypes the pooled count falls below the wild-only one. Projection returns the expected spectrum of a random subsample of 32 of the 52 haplotypes, and any site whose minor allele is absent from that subsample moves into the masked monomorphic corner; since a random subsample of the pooled set includes the less diverse captive animals, it holds fewer segregating sites on average than the 32 wild-caught haplotypes do. Projection preserves the expected shape of the spectrum, so the remaining rows of the table are unaffected.

Pooling shifts mass from the rarest bins of the spectrum towards the middle (Städler et al., 2009): bins 8–16 hold 26% more of the total, and Tajima’s D is 0.19 units less negative, whereas F_{ST} between the Scottish and domestic samples changes by only 2%. Since dadi cannot represent structure within a population, this excess would instead be interpreted as demography. Sampling across subpopulations mimics a recent change in population size (Chikhi et al., 2010; Mazet et al., 2016), and mass at intermediate frequencies informs the migration rates, so the parameters most at risk are the recent size and time of change in the wildcat branch and the rate of gene flow into it. All fits reported below therefore use only the wild-caught animals, haplotype columns 0–31.

2.3 Spectrum construction

The spectrum was folded because no outgroup is available for these data. It was built with `Spectrum.from_data_dict` using `polarized=False`, with `pop1 = SCO_WILD` and `pop2 = DOM`, at the full sample size of each population, giving a folded spectrum of shape (33, 13). The corner entries, which hold sites that are monomorphic in the sample and would otherwise dominate the likelihood through sheer count, were masked; under folding the (0, 0) and (n_1, n_2) bins are combined into one. Folding also masks the 208 redundant entries above the folding diagonal $i + j = 22$, whose counts are already carried by their reflected partners. The (n_1, n_2) corner lies above this diagonal and is one of those 208, so 209 entries are masked in total and the likelihood is evaluated over the remaining 220 of the 429 bins.

The spectrum contains 123,348 of the 145,512 biallelic sites. The remaining sites are polymorphic across the full set of 92 haplotypes but monomorphic within the 32 Scottish and 12 domestic haplotypes analysed here. All three models were fitted to this same spectrum (Figure 1), so their log-likelihoods are directly comparable.

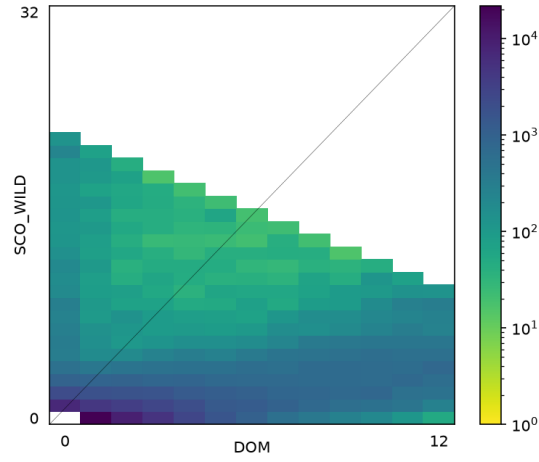


Figure 1: The folded joint site frequency spectrum used for all fits. It contains 123,348 segregating sites at the full sample size of each population (32 Scottish and 12 domestic haplotypes), giving a folded shape of (33, 13). Population order is `pop1 = SCO_WILD`, `pop2 = DOM`; axes are as described in Figure 3.

2.4 Models

Three models were fitted; their parameters are listed in Table 3 and their structure is shown in Figure 2. All three share the same topology, in which a single ancestral population splits into a wildcat branch and a domestic branch that then exchange migrants up to the present. They differ in how much structure is imposed on the population sizes and times, and in whether their parameters are intended to correspond to particular historical events.

Table 3: Parameters of the three models. The notation follows dadi for **sec_contact** and the original model specification for **basic** and **growth**; the two conventions are not interchangeable. All population sizes are fitted as ratios to the ancestral size N_A , which is fixed at 1 and absorbed into θ rather than being a free parameter; the corresponding size in individuals is the fitted ratio multiplied by N_{ref} (Section 2.7). The three times in **basic** and **growth** are measured backwards from the present, whereas T_1 and T_2 in **sec_contact** are epoch durations. Migration subscripts name the receiving population first in **sec_contact** and the source population first in the other two models; in all three the fitted quantity is the scaled rate $2N_A m$, despite the lower-case symbol. A leading subscript 2 denotes the second, more recent, migration epoch.

Model	Parameter	Interpretation
sec_contact (6 parameters)		
	ν_1	Scottish population after the split
	ν_2	Domestic population after the split
	m_{12}	Gene flow into Scottish from domestic
	m_{21}	Gene flow into domestic from Scottish
	T_1	Duration of isolation
	T_2	Duration of secondary contact
basic (11 parameters)		
	N_S	Silvestris branch after the split
	N_L	Lybica branch after the split
	N_B	Scottish wildcat from T_B to the present
	N_D	Domestic cat from T_D to the present
	T_A	Split of silvestris and lybica
	T_B	Size change in the wildcat branch
	T_D	Size change in the domestic branch; domestication
	m_{ls}	Lineage in silvestris came from lybica; T_A to T_B
	m_{sl}	Lineage in lybica came from silvestris; T_A to T_D
	$m_{2,ds}$	Lineage in wildcat came from domestic; T_B to the present
	$m_{2,sd}$	Lineage in domestic came from wildcat; T_D to the present
growth (13 parameters)		
	All eleven parameters of basic . N_B and N_D are the sizes reached at T_B and T_D rather than constants held from those times, and N_S and N_L are correspondingly the sizes at T_A rather than epoch constants. Plus:	
	$N_{B\text{curr}}$	Scottish wildcat at the present
	$N_{D\text{curr}}$	Domestic cat at the present

sec_contact (**sec_contact_asym_mig**) is distributed with dadi as part of the model set of Portik et al. (2017). It has six parameters, none of which is tied to a particular historical event, and it serves as the unlabelled comparator for Aim 1.

basic (**wildcat_domestic**) was specified by Mark Beaumont and implemented in dadi by Dennis Prangle, who used a large language model to translate the written specification into a model function and then checked the result by hand. Its two size changes are placed at named events in the history of the two species: the domestic branch changes size at T_D , representing domestication, and the wildcat branch at T_B , representing the loss of the British land bridge. The migration rates also change at these points, so that early and recent gene flow are separate parameters.

growth (**wildcat_domestic_growth**) retains the topology and migration structure of **basic** but

replaces the two instantaneous size changes with continuous exponential change. This adds two parameters and alters the meaning of T_B and T_D : in **basic** they are the times at which the size jumps, whereas in **growth** they are the times at which the rate of change alters.

A branch that changes exponentially from size N_0 at the start of an epoch to N_F at its end, over a duration T , has size $N(t) = N_0 (N_F/N_0)^{t/T}$, where t is measured forward from the start of the epoch. In **growth** the wildcat branch changes from N_B at T_B to $N_{B\text{curr}}$ at the present, and the domestic branch from N_D at T_D to $N_{D\text{curr}}$.

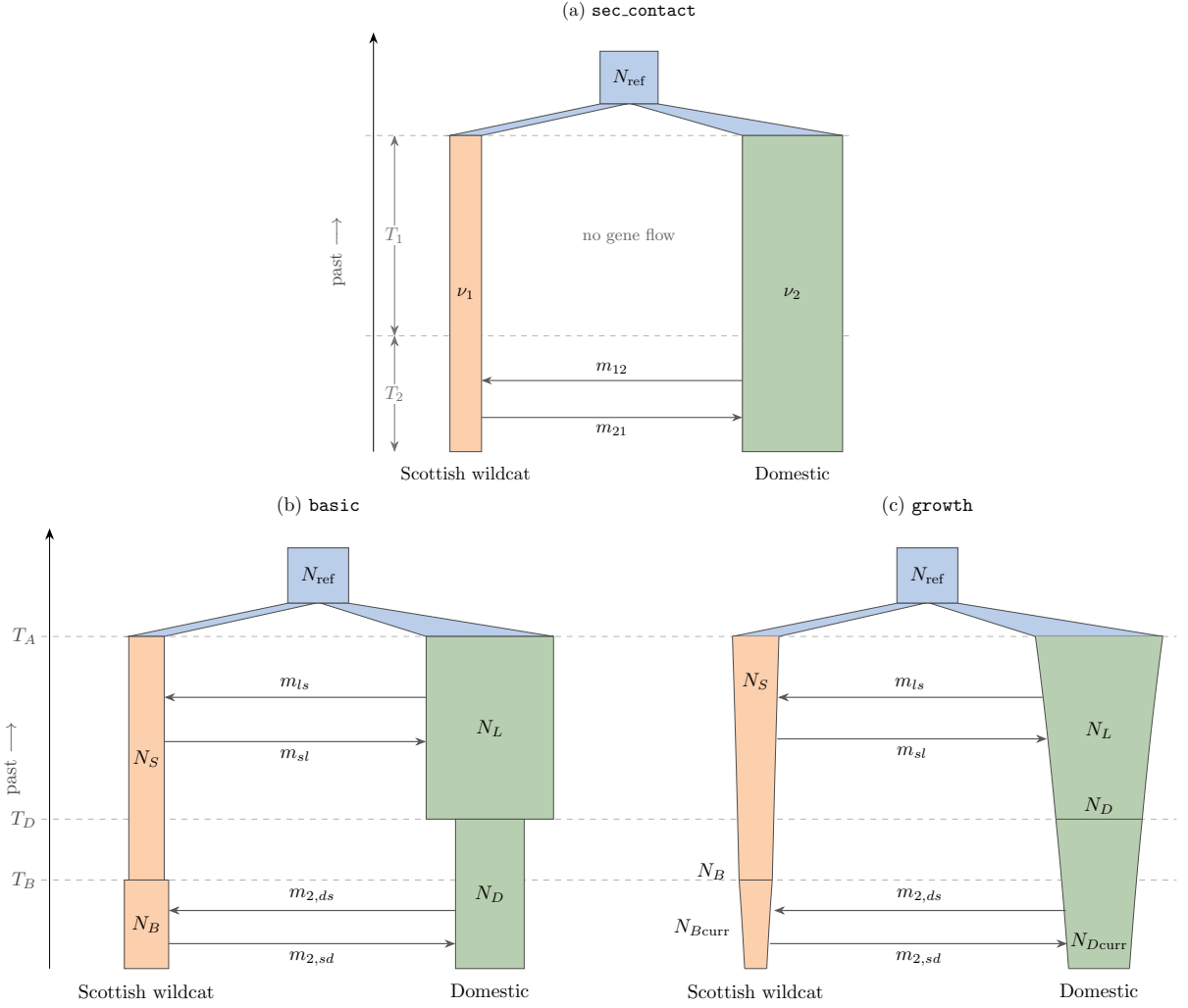


Figure 2: The three two-population models, with parameters as defined in Table 3 and scalings as in Section 2.7. (a) **sec_contact**: the ancestral population splits into two populations that remain isolated for a duration T_1 and then exchange migrants for a duration T_2 , so the total divergence time is $T_1 + T_2$. (b) **basic**: the split at T_A is followed by an instantaneous size change in each branch, at T_B in the wildcat branch and T_D in the domestic branch. (c) **growth**: the same topology and migration structure as (b), but with exponential rather than instantaneous size change. Panels (b) and (c) share a time axis. In (b) and (c) each direction of migration changes rate at the break point of the receiving branch, so that gene flow into the wildcat changes at T_B and gene flow into the domestic cat at T_D . Box widths indicate relative population size. Neither axis is to scale, and (a) is not drawn to the same scale as (b) and (c).

growth was fitted because the size changes in **basic** are instantaneous, whereas a real population

that changed size is more likely to have done so gradually. When the true change is gradual, the constant size that a single-epoch model fits to it approximates the harmonic mean of the underlying trajectory (Wright, 1938; Crow and Kimura, 1970), which is weighted towards the smaller values. The epoch sizes returned by **basic** therefore need not correspond to the size of the population at any particular point within the epoch. **growth** replaces each instantaneous change with a continuous exponential one at the cost of two additional parameters, so the comparison between the two models tests whether the data can distinguish these alternatives.

basic is not a special case of **growth**. In **growth** the population sizes are continuous at T_B and T_D by construction, so no choice of growth rates can reproduce the discontinuity that **basic** places at those times. The two models are therefore non-nested, which is why they are compared using an information criterion rather than a likelihood ratio test.

2.5 Optimisation

All models were fitted using dadi version 2.4.4 (Gutenkunst et al., 2009), on the same spectrum object and with the same grid points, `pts=[42, 52, 62]`. All fitting was carried out on the computational facilities of the Advanced Computing Research Centre, University of Bristol (<http://www.bristol.ac.uk/acrc/>).

basic and **growth** include the constraint $T_A > \max(T_B, T_D)$, and constrained optimisation in dadi is carried out by COBYLA through NLOpt. **sec_contact**, which has no such constraint, was fitted with `optimize_log_fmin`. The two routines use different stopping criteria, but this does not undermine the comparison, because convergence was assessed for each model in the same way, from the spread of log-likelihoods across its best restarts.

Four rounds of 50 restarts were run for each model. Each round was seeded from the best point found in the previous round, with the perturbation fold reduced from 2 to 1, 0.5 and 0.25 across the four rounds, so that early rounds searched widely and later rounds refined. Each restart was run as a separate SLURM array task, and rounds were chained with **afterany** rather than **afterok** so that a single task reaching its walltime would not terminate the chain. This schedule follows the structure of **dadi_pipeline** (Portik et al., 2017). All three models completed all four rounds.

The parameter bounds are stated here because three parameters finished at or close to a bound (Sections 3.2 and 4.7). In **basic** and **growth**, population sizes were bounded between 10^{-4} or 10^{-3} and 50, except for the two wildcat sizes N_B and $N_{B_{\text{curr}}}$, which were capped at 10 (a lower cap that did not bind, since neither fitted value nor confidence limit exceeded 0.5); T_A was bounded at 100 and T_B and T_D at 5; and all four migration rates were bounded at 20, the upper limit suggested in the dadi documentation on the grounds that larger values make evaluation slow. In **sec_contact**, sizes were bounded at 100, both times at 20 and both migration rates at 50.

The lower bounds on the migration rates in **basic** and **growth** were originally set to zero. However, COBYLA raises a **RoundoffLimited** exception when a parameter is pressed against a zero bound, so the lower bounds were raised to 10^{-4} on the scaled rate. At the fitted $N_{\text{ref}} \approx 1.35 \times 10^4$ this corresponds to a per-generation migrant fraction of $m \approx 3.7 \times 10^{-9}$, which is small enough that the models effectively retain the absence of gene flow where it is specified.

2.6 Model comparison and uncertainty

dadi maximises a composite likelihood, in which linked sites are treated as if they were independent. This overstates the amount of information in the data, and as a result the usual model-selection criteria are too generous towards complex models (Coffman et al., 2016). Since the three models have six, eleven and thirteen parameters, a correction was required before they could be compared.

Both the confidence intervals and the effective number of parameters were obtained from the Godambe information matrix, $G = HJ^{-1}H$, where H is the sensitivity matrix and J the variability matrix (Godambe, 1960; Varin et al., 2011). H was estimated at the maximum-likelihood point $\hat{\Theta}$ by central finite differences, and J as the covariance of the score across approximately independent blocks of the genome.

Models are compared by

$$\text{CLAIC} = -2\ell_{\text{CL}}(\hat{\Theta}) + 2\text{tr}(JH^{-1}), \quad (1)$$

where $\text{tr}(JH^{-1})$ is the effective number of parameters. This is the composite-likelihood AIC of Varin and Vidoni (2005). dadi version 2.4.4 does not implement CLAIC, so J and H were extracted as intermediate quantities from `Godambe.get_godambe` and combined directly. Later versions of dadi provide `Godambe.CLAIC`, which writes the penalty as $\text{tr}(HG^{-1})$; since $G = HJ^{-1}H$, this is identical to the $\text{tr}(JH^{-1})$ of Equation 1.

J was estimated from 100 block-bootstrap replicates over 1 Mb blocks generated by `Misc.fragment_data_dict`, which divides the data by physical position rather than by callable sites, giving roughly 415 blocks across the 414 Mb spanned by the two chromosomes. The limiting factor is therefore the number of replicates rather than the number of blocks. A covariance matrix estimated from n replicates has sampling error of order $1/\sqrt{n}$ in its entries, so with 100 replicates J , and hence $\text{tr}(JH^{-1})$, would be expected to carry noise of roughly 10–15%. This figure was not measured directly, for instance by repeating the bootstrap with different seeds, and is an order-of-magnitude expectation rather than an estimate. Dividing a single genome into sub-regions that are treated as approximately independent is the window subsampling approach of Heagerty and Lumley (2000), which requires that blocks be long relative to the scale of linkage disequilibrium.

Confidence intervals were obtained from the same J and H , using the sandwich covariance $G^{-1} = H^{-1}JH^{-1}$. dadi’s own `Godambe.GIM_uncert` was not used, for two reasons. First, it recomputes both matrices from the bootstrap replicates rather than reusing those underlying the CLAIC, so its intervals would not refer to the same J and H . Second, it forms $G = HJ^{-1}H$ and then inverts G , which requires two matrix inversions, whereas writing the covariance directly as $H^{-1}JH^{-1}$ requires inverting only H . This matters for a matrix whose condition number is of the order of 10^{15} .

Standard errors were converted to the log scale by the delta method (Oehlert, 1992), $\sigma_{\log p} = \sigma_p/\hat{p}$, so that the resulting intervals remain positive; on the linear scale, for example, the lower limit for T_B is negative, which is not a meaningful time. The intervals reported are therefore Wald intervals on $\log p$, exponentiated. They are multiplicative about $\hat{p}$ and span a factor of $\exp(3.92\sigma_{\log p})$, which is how their widths are described below. Derivatives were taken on the linear scale throughout, because in log space dadi’s step-size rule reverts to a one-sided difference for every parameter below one, and on these data that introduces a negative eigenvalue into H . Rescaling G^{-1} to a correlation matrix, $R_{ij} = C_{ij}/\sqrt{C_{ii}C_{jj}}$, gives the correlations reported in

Section 3.6. The intervals, the CLAIC and the correlations therefore all derive from the same pair of matrices at the same step size, and no model was refitted for any of them.

All three models were evaluated at two finite-difference step sizes, $\varepsilon = 0.01$ and 0.005 . A large change in $\text{tr}(JH^{-1})$ between step sizes would indicate that the finite differences have not converged, and a negative smallest eigenvalue of H would indicate that the fit is not at an interior maximum; both quantities are reported. It should be noted that the intervals and the CLAIC both assume an interior maximum. At a parameter bound the usual asymptotic results for maximum likelihood estimators do not hold (Chernoff, 1954; Self and Liang, 1987), and the finite differences are taken about a point that is not a maximum at all. All intervals reported below are at $\varepsilon = 0.01$.

Residuals are reported as Anscombe residuals, which normalise the difference between model and data by a variance-stabilising transformation so that entries with small expected counts are not dominated by Poisson noise:

$$residual = \frac{3}{2} \cdot \frac{(model^{2/3} - \frac{1}{9}model^{-1/3}) - (data^{2/3} - \frac{1}{9}data^{-1/3})}{model^{1/6}}. \quad (2)$$

The difference is taken as model minus data, so a positive residual indicates an entry in which the model predicts more segregating sites than were observed.

2.7 Conversion to physical units

Parameters were converted from coalescent to physical units using `dadi`'s scaling conventions. These differ from those of coalescent simulators such as `ms` by factors of two in several places, and so are stated explicitly here. The reference population size follows from the fitted θ as

$$\theta = 4N_A\mu L \quad \implies \quad N_{\text{ref}} = \frac{\theta}{4\mu L}, \quad (3)$$

where N_A is the size of the ancestral population. Every other size is fitted relative to N_A , so $N_A = 1$ by construction and N_{ref} is its value in individuals; a fitted size x therefore corresponds to xN_{ref} individuals. These ratios are written ν_i in `sec_contact`, and as N with a subscript naming the branch in `basic` and `growth`. Times are expressed in units of $2N_{\text{ref}}$ generations, so that

$$T_{\text{gen}} = 2N_{\text{ref}}\tau, \quad T_{\text{yr}} = T_{\text{gen}} \times g, \quad g = 3 \text{ yr}. \quad (4)$$

Migration rates are scaled as $M = 2N_A m$, where m is the fraction of the receiving population that is made up of new migrants from the source population each generation. Tables 3, 5 and 6 label these parameters m , following the original model specification, but the quantity actually fitted and reported is the scaled rate M . The expected number of migrant individuals entering a population of fitted size ratio N each generation is $NM/2$.

The distinction between a rate and a count matters when interpreting asymmetric migration. Because the number of migrants scales with the size of the *receiving* population rather than with N_{ref} , a large asymmetry in rates can arise purely from a difference in population sizes. Both quantities are therefore reported together in the Results.

Each of the three models has its own N_{ref} , so the scaled parameters are not directly comparable between models: the same value of ν , τ or M corresponds to a different number of individuals,

years or migrants in each fit. Conversion to physical units removes this difficulty, and comparisons between models are therefore made on the physical values and on the per-generation rates.

3 Results

3.1 Model comparison

Fit statistics for the three models are given in Table 4. The **growth** model had the highest log-likelihood and the lowest CLAIC, with **basic** 34.07 units above it and **sec_contact** 840.37 units above it; relative to **basic**, **sec_contact** is 806.30 units higher. The **growth** fit terminated with N_S on its upper bound rather than at an interior maximum (Section 3.2).

The effective number of parameters, $\text{tr}(JH^{-1})$, ranged from 292.73 to 338.33 across the three models, against nominal parameter counts of 7, 12 and 14 including θ . The models rank in the same order under unadjusted AIC as under CLAIC.

Table 4: Fit statistics for the three models at a step size of $\varepsilon = 0.01$. The parameter count k includes θ . Differences in CLAIC are taken relative to **basic**. AIC is unadjusted and is shown for reference only. The **growth** fit terminated on a parameter bound rather than at a maximum; the implications of this for its CLAIC are discussed in Section 4.5.

Model	ℓ	k	Effective k	CLAIC	ΔCLAIC	AIC
basic	-1411.14	12	292.73	3407.74	0.00	2846.29
growth	-1376.22	14	310.62	3373.67	-34.07	2780.44
sec_contact	-1768.69	7	338.33	4214.04	+806.30	3551.37

3.2 Optimiser diagnostics

All 50 restarts in the final round returned a fit for every model (Appendix C, Table A1). The gap in log-likelihood between the best and tenth-best restart was 0.000 for **sec_contact**, 0.107 for **basic** and 0.903 for **growth**.

No parameter of **basic** finished on a bound; the closest was $m_{2,ds}$, at 18.76 against an upper bound of 20, with a confidence interval reaching 19.51. In **sec_contact**, m_{12} finished on its upper bound of 50, and in **growth**, N_S finished at 49.97 against an upper bound of 50. The confidence intervals for both of these parameters extend beyond the bound.

3.3 Parameter estimates

Parameter estimates, 95% confidence intervals and physical values are given in Tables 5 to 7. Intervals were computed on the log scale (Section 2.6) and are therefore asymmetric about the estimate. Notation, scaling and the migration conventions follow Table 3 and Section 2.7: migration parameters are the scaled rate $2N_A m$ in all three models, and migrant counts are

individuals per generation entering the receiving population, which depend on the size of that population rather than on N_{ref} .

Table 5: Parameter estimates for **basic** (log-likelihood -1411.14). The fitted $\theta = 10,454.38$ gives $N_{\text{ref}} = 13,514$. Sizes are ratios to $N_A = 1$, and the physical value is the estimate multiplied by N_{ref} . Migration parameters are scaled rates $2N_A m$; their physical value is the expected number of migrant individuals entering the receiving population per generation.

Parameter	Estimate	95% CI	Physical value
N_S	0.1542	0.06499–0.3661	2,084 individuals
N_L	3.4633	1.541–7.785	46,802 individuals
N_B	0.2006	0.1621–0.2482	2,711 individuals
N_D	1.6585	0.6409–4.292	22,413 individuals
T_A	5.9872	2.299–15.59	485,461 years
T_B	0.0083	0.007988–0.008631	673 years
T_D	0.1307	0.0395–0.4326	10,599 years
m_{ls}	0.3390	0.1343–0.8555	0.026 migrants/gen
m_{sl}	0.1399	0.06636–0.295	0.242 migrants/gen
$m_{2,ds}$	18.7571	18.03–19.51	1.881 migrants/gen
$m_{2,sd}$	6.0652	4.486–8.201	5.030 migrants/gen
θ	10,454.38	4,753–22,990	—

Four parameters of **basic** have confidence intervals that span a factor of two or less: N_B (a factor of 1.5), T_B (1.1), $m_{2,ds}$ (1.1) and $m_{2,sd}$ (1.8). The remaining intervals span factors of between 4.4 and 11.0, the widest being T_D at 11.0; T_A spans a factor of 6.8.

Table 6: Parameter estimates for **growth** (log-likelihood -1376.22). The fitted $\theta = 9,784.88$ gives $N_{\text{ref}} = 12,648$; scaling is as in Table 5. Migrant counts are evaluated at the size of the receiving population at one end of the epoch over which the rate applies: at T_A for m_{ls} and m_{sl} , and at the present for $m_{2,ds}$ and $m_{2,sd}$. Because population size changes continuously in this model, the count varies across each epoch, and the value given is not an average. N_S finished on its upper bound, and its confidence interval extends beyond that bound.

Parameter	Estimate	95% CI	Physical value
N_S	49.9665	4.352–573.7	631,994 individuals
N_L	0.9262	0.2395–3.582	11,714 individuals
N_B	0.1472	0.0414–0.5235	1,862 individuals
N_D	4.8539	1.139–20.68	61,393 individuals
T_A	6.0816	1.144–32.33	461,534 years
T_B	0.0112	0.009218–0.01364	851 years
T_D	0.5723	0.1498–2.187	43,434 years
m_{ls}	0.2775	0.1045–0.7371	6.933 migrants/gen
m_{sl}	0.1913	0.03981–0.9188	0.089 migrants/gen
$m_{2,ds}$	14.2695	11.4–17.86	2.359 migrants/gen
$m_{2,sd}$	3.6523	2.539–5.253	3.096 migrants/gen
$N_{B\text{curr}}$	0.3306	0.2213–0.4937	4,181 individuals
$N_{D\text{curr}}$	1.6952	0.572–5.024	21,441 individuals
θ	9,784.88	3,326–28,780	—

Two parameters of **growth** have intervals that span a factor of two or less: T_B (1.5) and $m_{2,ds}$ (1.6). Excluding N_S , the remaining intervals span factors of between 2.1 and 28.3, the widest being T_A .

Table 7: Parameter estimates for **sec_contact** (log-likelihood -1768.69). The fitted $\theta = 37,301.73$ gives $N_{\text{ref}} = 48,218$; scaling is as in Table 5. m_{12} finished on its upper bound of 50, and its confidence interval extends beyond that bound.

Parameter	Estimate	95% CI	Physical value
ν_1	0.0603	0.04737–0.07673	2,907 individuals
ν_2	0.6408	0.5735–0.7159	30,896 individuals
m_{12}	50.0000	31.75–78.75	1.507 migrants/gen
m_{21}	22.1165	15.77–31.02	7.086 migrants/gen
T_1	0.1603	0.1193–0.2155	46,389 years
T_2	0.0033	0.002071–0.005412	969 years
θ	37,301.73	35,380–39,330	—

The total divergence time under **sec_contact** is $T_1 + T_2 = 0.1636$, or 47,357 years. Excluding m_{12} , every interval spans a factor of 2.6 or less.

3.4 Effect of the step size on the Godambe correction

Both J and H are computed by finite differences with a step size ε , and the calculation was repeated at two step sizes to check that the results did not depend on this choice. Table 8 gives the results, together with the smallest eigenvalue and the condition number of H at each step size.

Halving the step size changed the effective parameter count by 0.56 for **basic**, 2.51 for **growth** and 21.15 for **sec_contact**, and the corresponding CLAIC values by 1.12, 5.01 and 42.31. The ranking of the three models was unchanged. The gap between **growth** and **basic** was 34.07 units at $\varepsilon = 0.01$ and 40.20 units at $\varepsilon = 0.005$, and the gap between **basic** and **sec_contact** was 806.30 and 762.87 units respectively.

At $\varepsilon = 0.01$ the smallest eigenvalue of H was positive for all three models, although no larger than 3.73×10^{-5} in any of them. At $\varepsilon = 0.005$ it remained positive for **sec_contact** and **growth** but was negative for **basic**. The condition number of H was of the order of 10^{14} for **sec_contact** and **growth**, and 10^{15} for **basic**, at both step sizes.

Table 8: Godambe quantities at two finite-difference step sizes, with the smallest eigenvalue and the condition number of H at each.

Model	ε	Effective k	CLAIC	Smallest eigenvalue	Condition number
sec_contact	0.01	338.33	4214.04	$+3.73 \times 10^{-5}$	1.7×10^{14}
sec_contact	0.005	317.18	4171.73	$+3.83 \times 10^{-5}$	1.7×10^{14}
basic	0.01	292.73	3407.74	$+6.33 \times 10^{-7}$	1.4×10^{15}
basic	0.005	293.29	3408.86	-9.24×10^{-1}	1.1×10^{15}
growth	0.01	310.62	3373.67	$+6.06 \times 10^{-7}$	8.2×10^{14}
growth	0.005	308.11	3368.66	$+8.65 \times 10^{-7}$	5.8×10^{14}

3.5 Fit to the spectrum

Figures 3 to 5 show the observed spectrum, the expected spectrum under each fitted model, and the Anscombe residuals.

The largest absolute residual was 7.7 for **basic**, 7.6 for **growth** and 11 for **sec_contact**. For **basic** and **growth** the residual histograms are centred close to zero, and the residual maps show no large regions of a single sign. For **sec_contact**, the largest residuals are concentrated in the bins where both the domestic and the wildcat allele counts are low, and the residual histogram contains values beyond ± 8 that do not occur for the other two models.

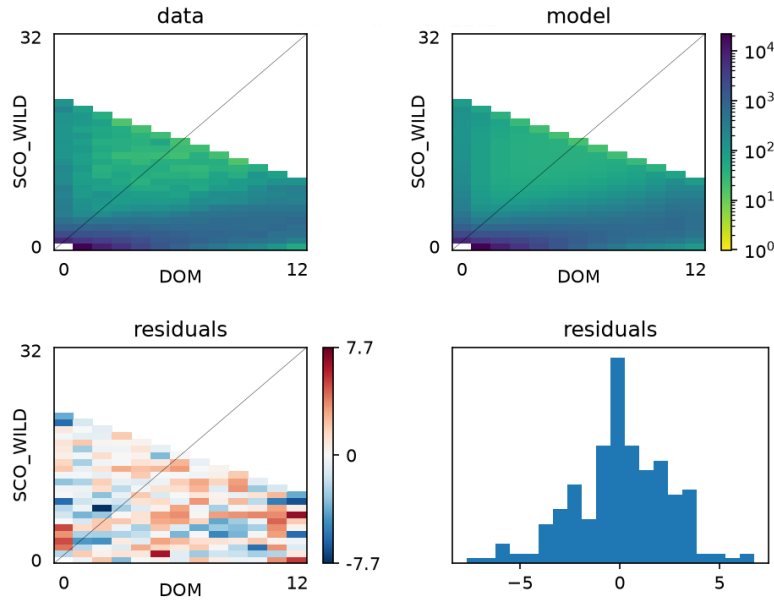


Figure 3: Fit of **basic** ($\ell = -1411.14$). Upper left: the observed spectrum. Upper right: the expected spectrum under the fitted model. Each entry is coloured by the logarithm of the number of sites it contains, and masked entries are shown in white. Because the spectrum is folded, entry (i, j) counts sites at which the allele that is minor across the pooled 44 haplotypes has i copies among the 32 Scottish wildcat haplotypes (vertical axis) and j copies among the 12 domestic haplotypes (horizontal axis); neither axis is a derived-allele count. Lower panels: the Anscombe residuals and their histogram. The colour scale is symmetric and spans ± 7.7 .

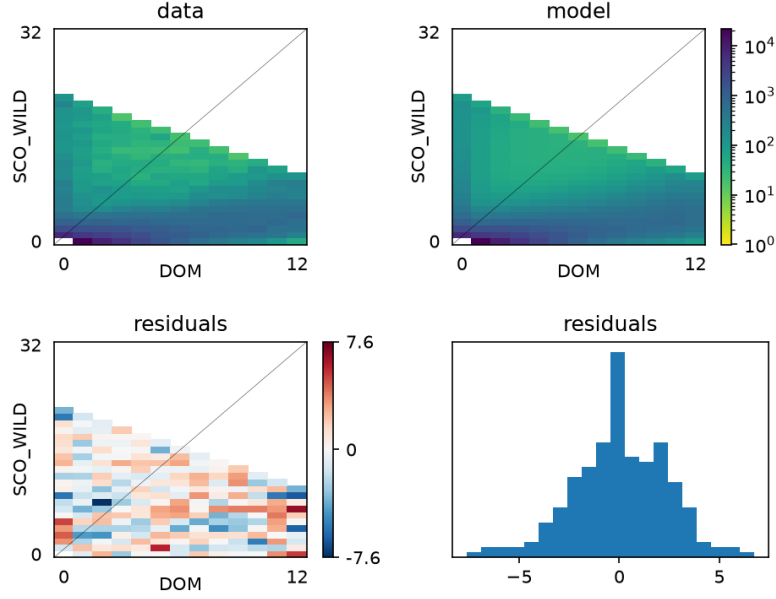


Figure 4: Fit of **growth** ($\ell = -1376.22$). Panels and axes are as in Figure 3. The colour scale of the residual map is symmetric and spans ± 7.6 . The model shown is a constrained optimum, since N_S finished on its upper bound.

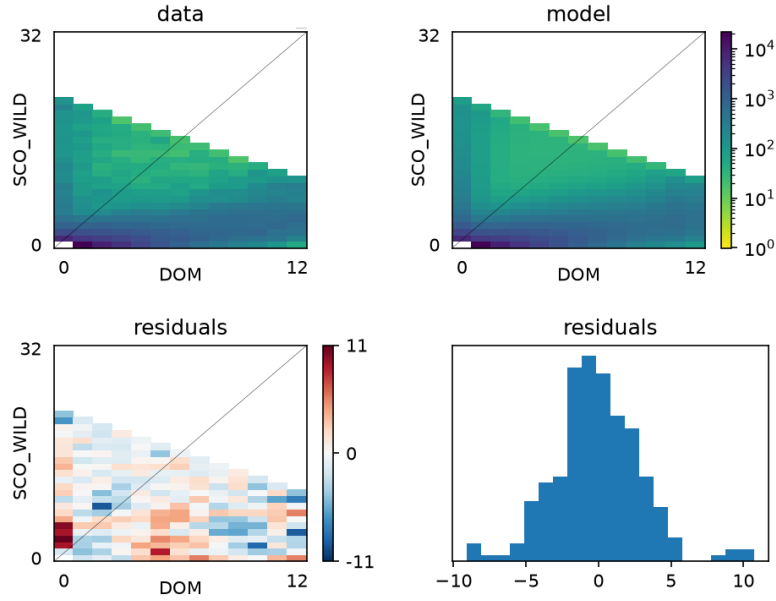


Figure 5: Fit of **sec_contact** ($\ell = -1768.69$). Panels and axes are as in Figure 3. The colour scale of the residual map is symmetric and spans ± 11 , which is wider than for the other two fits, so residual colours are not directly comparable between figures.

3.6 Parameter correlations

Correlation matrices derived from the sandwich covariance at $\varepsilon = 0.01$ are shown for **basic** in Figure 6 and for the other two models in Appendix D (Figures A2 and A3). The values reported below are all pairs whose correlation exceeds 0.95 in absolute value, together with the strongest

pairs below that threshold where these are needed to place a parameter.

In **basic**, sixteen pairs exceed 0.95. N_S , N_L and T_A correlate with one another at between +0.975 and +0.983, and each correlates with θ at -0.990 or below. N_D belongs to the same group, correlating with N_L at +0.954, with T_A at +0.953, with θ at -0.953 and with N_S at +0.948. The early migration rates correlate negatively with these sizes and positively with θ : m_{ls} with N_S at -0.974 , with N_L at -0.954 and with θ at +0.956; m_{sl} with N_L at -0.971 , with N_S at -0.963 and with θ at +0.969. Separately from this group, T_B and $m_{2,ds}$ correlate at +0.987. The strongest correlation involving T_D is +0.911, with N_D ; the strongest involving N_B is 0.488 and the strongest involving $m_{2,sd}$ is 0.193.

In **growth**, fifteen pairs exceed 0.95. N_B , N_D and T_A correlate with one another at between +0.976 and +0.988, each correlates with θ (at -0.996 , -0.972 and -0.987) and with m_{sl} (at -0.983 , -0.986 and -0.994), and m_{sl} correlates with θ at +0.981. N_{Dcurr} joins this group at -0.966 with θ and +0.961 with N_B , and m_{ls} correlates with N_B at -0.952 . N_S falls outside the group: its two strong correlations are with $m_{2,ds}$ at +0.983 and with T_B at -0.980 . T_B and $m_{2,ds}$ correlate at -0.931 , the opposite sign to **basic**. The strongest correlation involving T_D is +0.936, with N_{Dcurr} ; for N_{Bcurr} , N_L and $m_{2,sd}$ the strongest are 0.650, 0.578 and 0.342.

In **sec_contact**, only two pairs exceed 0.95: m_{12} with T_2 at -0.990 , and ν_1 with T_1 at +0.958. Below that, m_{12} correlates with T_1 at -0.926 and with ν_1 at -0.925 , T_1 with T_2 at +0.912, ν_1 with T_2 at +0.907, and the two migration rates with each other at +0.823. ν_2 correlates positively with ν_1 , T_1 and T_2 (at +0.710 to +0.756) and negatively with both migration rates and with θ . The strongest correlation involving θ is 0.563, with ν_1 .

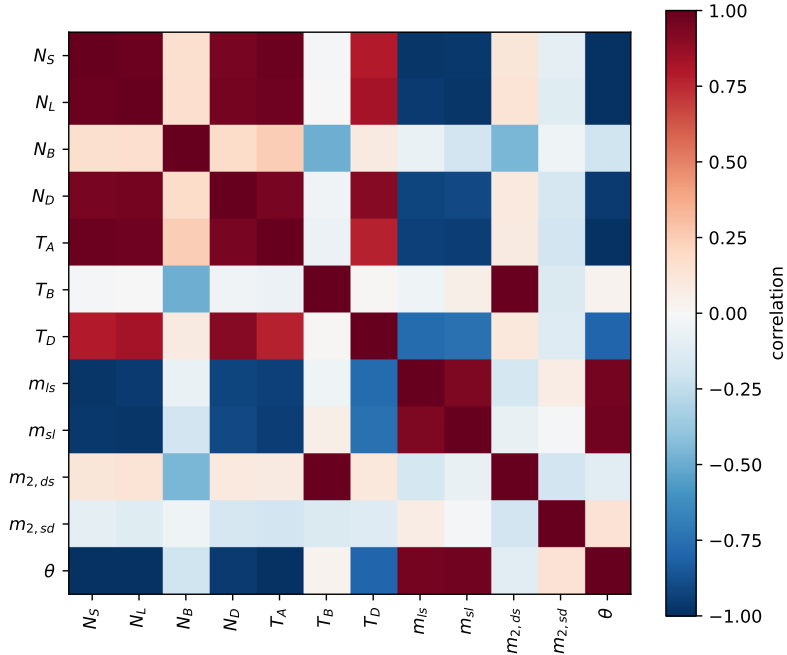


Figure 6: Correlation matrix for **basic** at $\varepsilon = 0.01$, obtained by rescaling the sandwich covariance $G^{-1} = H^{-1}JH^{-1}$ whose diagonal gives the intervals in Table 5. θ is the last row and column, since `dadi` appends it after the model parameters when the matrices are computed with `multinom=True`. Red is positive, blue negative, and the diagonal is one by construction. Because H has a condition number of around 10^{15} , values near ± 1 should be read as indicating that two parameters are not separately identified, rather than as precise measures of covariation.

4 Discussion

4.1 What the model comparison establishes

sec_contact scores 806 CLAIC units worse than **basic**, roughly twenty times the 40-unit variation in the statistic across finite-difference step sizes. What separates it from the other two models is structure: **basic** and **growth** both allow early and recent gene flow to differ and allow each branch to change size independently, whereas **sec_contact** has a single size per branch and a single period of contact. Since the optimiser knows nothing of the historical labels attached to the parameters, Aim 1 is answered affirmatively only in a narrow sense: the spectrum favours more structure than **sec_contact** allows, but it does not by itself identify the historical events.

sec_contact also finished with m_{12} on a bound, so its CLAIC is open to the same objection that is raised against **growth** in Section 4.5. The objection carries less weight here, for three reasons. The gap is 806 units rather than 34, an order of magnitude beyond both the step-size variation and the noise in the penalty; the parameter affected is one migration rate out of six, rather than the size parameter that drives the fit; and the misfit is visible in the residual map independently of the statistic. The direction of the comparison is therefore secure, even though the number attached to it is not exact.

When a model is misspecified, J and H diverge, and the sandwich penalty then reflects misfit as well as the number of parameters (Varin et al., 2011). **sec_contact** carries the largest penalty despite having the fewest parameters, which reflects misfit rather than complexity, and it is also the model least stable to the choice of step size. Its two migration rates, at 50.0 and 22.1, are the highest fitted in any of the three models, and both are at or above 20, the value given in the dadi documentation as an upper bound for migration parameters on the grounds that larger values make evaluation slow. No published analysis quantifies how the accuracy of the diffusion approximation degrades as migration rates rise, so this is a caution rather than a diagnosis, but part of the misfit may be numerical. The recent rate into the wildcat approaches the same value in **basic** and **growth**, at 18.8 and 14.3, so the caveat applies to those estimates as well; no grid-refinement check was run for any of the three models.

No model reproduces the spectrum fully. The largest Anscombe residual is 11 for **sec_contact**, against 7.7 and 7.6 for **basic** and **growth**. A rough adjustment for the composite-likelihood inflation, dividing each residual by the square root of the ratio of effective to nominal parameter count, brings all three to between 1.56 and 1.61, which suggests that the models cannot be distinguished by the magnitude of their residuals alone. Their pattern is another matter. The largest residuals of **sec_contact** are concentrated in the bins where both populations have low allele counts, whereas those of **basic** and **growth** show no regions of a single sign, and structure of that kind is precisely what a residual map should not show if the model were adequate.

sec_contact is also the model whose parameters are least separately identified, despite having the narrowest intervals of the three. Every interval except that of m_{12} spans a factor of 2.6 or less, yet m_{12} and T_2 correlate at -0.990 and the six parameters fall into two blocks of opposite sign (Figure A3). Narrow marginal intervals and joint degeneracy are not in conflict, but only the first is visible in Table 7.

4.2 Can the historical labels be tested?

The estimate of T_D at 10,600 years coincides with the earliest evidence of cats accompanying humans on Cyprus, at a settlement dated to at least 10,600 years before present (Vigne et al., 2012). Cats are not native to the island, so their presence implies transport from the mainland and a commensal relationship already established in the eastern Mediterranean. This evidence does not, however, date the emergence of a genetically distinct domestic population, and the fitted parameter is a size change in the *lybica* branch, which need not coincide with either event. The match is in any case only as good as the confidence interval of 3,200–35,100 years allows, and that interval is too wide for the estimate to be treated as a date.

T_A is difficult to verify, because published estimates disagree with one another and none of them measures the population split directly. Driscoll et al. (2007) date the mitochondrial coalescence of the *F. silvestris* complex, including *F. s. bieti*, to around 230,000 years from a single mitochondrial region, whereas Johnson et al. (2006) used nuclear DNA to place the corresponding divergence at around 1.4 million years. Both are gene divergence times, which necessarily exceed the population split time, since lineages sampled from two populations can only coalesce in their common ancestor. The interval of 186,000–1,264,000 years spans the mitochondrial estimate and reaches to within 10% of the nuclear one, and naively propagating the uncertainty in N_{ref} , whose own interval spans a factor of five, would widen it to cover both. The estimate is therefore consistent with the published figures, but it cannot be checked against them.

T_B corresponds to none of the candidate events. The final inundation of Doggerland is dated to roughly 8,200 calibrated years before present (Weninger et al., 2008), more than ten times older than the fitted 673 years, whereas the persecution-driven decline of the nineteenth century (Langley and Yalden, 1977) and the recent genetic swamping of the Scottish population (Howard-McCombe et al., 2023) are both more recent. T_B has the narrowest confidence interval of any parameter in **basic**, at 648 to 700 years, and even naively propagating the uncertainty in N_{ref} widens it only to roughly 300 to 1,500 years. The most likely explanation is that the wildcat branch has a single break point where the true history contains several candidate events, so that the fitted time settles at a compromise matching none of them. The land-bridge label should therefore be dropped; there is no reason to expect a refit to recover it.

4.3 What the data determines

Turning to Aim 2, four parameters of **basic** are determined to within a factor of two: N_B , T_B , $m_{2,ds}$ and $m_{2,sd}$. Three of these describe the epoch since T_B ; the fourth, $m_{2,sd}$, applies from T_D . No other parameter of **basic** is determined to better than a factor of four.

Recency alone does not account for this. N_D also describes a recent epoch, that of the domestic branch from T_D to the present, yet its interval spans a factor of 6.7, and its epoch is not shorter than that of N_B in coalescent units. The more likely explanation is sample size: the domestic sample is roughly a third the size of the Scottish one, at 12 haplotypes against 32, which leaves fewer lineages to coalesce and gives a marginal spectrum of 13 bins rather than 33. Being recent is therefore necessary but not sufficient for a parameter to be well determined here.

Agreement between models is not evidence that a parameter is well determined. **basic** and **growth** return values of N_{ref} within 7% of each other and of T_A within 5%, yet within **basic** the intervals on those two quantities span factors of 4.8 and 6.8. Two parameterisations can return

the same value without either one being well determined.

The correlation matrix shows that these interval widths are not independent of one another. N_S , N_L and T_A , whose intervals span factors of 5.6, 5.1 and 6.8, correlate with each other at above +0.97 and with θ at below -0.99, and all three correlate negatively with the early migration rates. They are not three separately uncertain quantities but points along a single ridge in the likelihood: increasing the ancestral sizes and lengthening the time since the split, while decreasing θ and the early gene flow, leaves the composite likelihood almost unchanged. Models of many parameters that are tightly constrained along a few directions and nearly flat along the rest are common enough to have acquired a name, “sloppy” models, and the eigenvalue spectrum of H here has the shape typical of them (Gutenkunst et al., 2007; Transtrum et al., 2015). This is also why **basic** and **growth** can return N_{ref} and T_A within a few percent of each other without either being well determined: both fits are free to move along the same ridge.

The narrow intervals are correspondingly less informative than they appear. T_B and $m_{2,ds}$, two of the four parameters determined to within a factor of two, correlate at +0.987. What the data fix is a combination of the length of the recent epoch and the rate of gene flow within it, rather than either quantity separately, so the interval of 648 to 700 years on T_B holds only at the fitted value of $m_{2,ds}$. Of the four narrow parameters, only N_B and $m_{2,sd}$ are free of this: neither correlates with any other parameter above 0.49 (Section 3.6).

Finally, **basic** places T_D at 10,600 years and **growth** at 43,400 years, and each interval excludes the other model’s estimate: 3,200 to 35,100 years against 11,400 to 166,000 years. Two models that the data cannot separate therefore give incompatible answers for the same event. This is model uncertainty, which none of the intervals reported here includes, so the true uncertainty is wider than the tables suggest.

4.4 Introgression into the Scottish population

Both **basic** and **growth** place recent gene flow from the domestic cat into the wildcat at a high rate relative to the other three migration parameters, and this asymmetry holds across two models that disagree about almost every ancestral quantity. The migrant counts run the other way, at 5.03 individuals per generation into the domestic population against 1.88 into the wildcat, because the receiving populations differ in size by a factor of eight. Rate and count therefore rank the two directions oppositely, and neither ordering is more correct than the other.

Comparison with the literature requires care, because the published figures are different quantities. Howard-McCombe et al. (2021) estimate a per-generation migration rate of 13% from an approximate Bayesian computation model, applied over a short and recent period of hybridisation; this is a rate. Howard-McCombe et al. (2023) report domestic ancestry of 11 to 74% in modern Scottish individuals; this is the cumulative result of migration over many generations.

Compounding the rate fitted here over the 224 generations from T_B to the present, as $1 - (1 - m)^t$, gives 14.4% domestic ancestry, which is at the lower end of the observed range of 11 to 74%. The two rates are not directly comparable, since the 13% of Howard-McCombe et al. (2021) is applied over a few recent generations rather than over the whole of the recent epoch, and compounded over 224 generations it would imply almost complete replacement. Rather, the two describe different histories that reach similar cumulative ancestry: a low constant rate over a long period here, and a high rate over a short one there. This suggests a hypothesis, that

the spectrum constrains cumulative introgression more tightly than it constrains the rate and duration separately. The agreement is with a wide range rather than a particular value, and it rests on a duration that the model itself estimates, so it is offered as no more than that. What a constant rate cannot reproduce is the spread of ancestry among individuals. It predicts the same expected ancestry for every animal, whereas the observed values span a factor of six, and it distributes that ancestry evenly across the epoch rather than concentrating it in recent generations, which is what Howard-McCombe et al. (2023) find.

The correlation of $+0.987$ between T_B and $m_{2,ds}$ in **basic** (Section 3.6) is consistent with this hypothesis. The duration and the rate are constrained jointly rather than separately, which is what would be expected if a cumulative quantity were determined more tightly than either of its factors.

4.5 The growth fit

growth has the lowest CLAIC, but the fit behind that number is not a maximum. N_S finished at 49.97, just below its upper bound of 50, having increased in every round of optimisation (Table A2). The log-likelihood was still improving, by 6.5 units in the final round and by a further 0.574 units when the fit was polished, which indicates that the optimiser stopped because it reached the bound rather than because it found an optimum.

CLAIC assumes that the fit is at a maximum, where the likelihood falls away in every direction and H measures how sharply it does so. There is no such point here, and the asymptotic theory on which the statistic rests does not hold at a boundary (Chernoff, 1954; Self and Liang, 1987). The finite differences used to compute H also step above 50, outside the range the optimiser was permitted to search. The 34-unit advantage over **basic** therefore does not demonstrate that **growth** fits better.

The confidence interval on N_S runs from 4.35 to 573.7, a factor of 132. The likelihood therefore changes very little as N_S varies, and the optimiser continues to move until a bound stops it. A large early silvestris size implies very little drift in that epoch, and drift in the oldest epoch is precisely what Section 4.3 shows these data constrain least.

The correlation matrix shows the same thing in a third way. The twelve strongest correlations in **growth** all exceed 0.97 in absolute value, against a twelfth-strongest of 0.956 in **basic** and 0.710 in **sec_contact**, and N_S itself correlates with $m_{2,ds}$ at $+0.983$ and with T_B at -0.980 . A parameter that trades off so strongly against two others is one that the likelihood barely distinguishes, which is what drives it to a bound rather than to an optimum.

The expectation set out in Section 2.4 was met only in part. The early silvestris size rose under **growth**, as the harmonic-mean argument of Wright (1938) implies it should, but the early lybica size fell. The two models are not, however, describing the same epoch: **growth** places T_D at 43,400 years and **basic** at 10,600 years, so the sizes being compared cover different spans of time.

basic is therefore reported as the primary result. It is the only model with no parameter on a bound, and its effective parameter count is the least sensitive to the step size, changing by 0.56 against 2.51 for **growth** and 21.15 for **sec_contact**. Its smallest eigenvalue of H does become negative at $\varepsilon = 0.005$, but this most likely reflects a flat direction in the likelihood rather than

a fit away from a maximum, since the effective parameter count barely changes between step sizes, and a condition number of 10^{15} leaves almost no significant digits in double precision with which to resolve the sign (Higham, 2002). In any case the two models could not be separated. If each penalty carries the roughly 10–15% noise expected from 100 replicates (Section 2.6), that is ± 60 to ± 90 units on penalties near 600, and a 34-unit gap is smaller than the correction can be expected to resolve. Repeating the bootstrap with different seeds would put a measured figure on this.

4.6 The baseline for simulation-based inference

Aim 3 asks for a baseline rather than a conclusion, so what matters here is which quantities can safely be compared against and which cannot. The point estimates and interval widths in Tables 5 to 7 are the obvious targets, but three features of them qualify any such comparison. The argument that follows is not drawn from the literature: simulation-based inference is well described in general terms (Cranmer et al., 2020), but no published work directly addresses how its uncertainty estimates should be compared with those of a composite likelihood.

First, the intervals are marginal. Section 4.3 shows that the deep parameters of **basic** lie along a single ridge, and that T_B and $m_{2,ds}$ are constrained only in combination. A method that recovered the same marginal widths while placing its probability mass along a different ridge would appear to agree in a table while disagreeing in substance. Figure 6 is therefore part of the baseline itself, not merely a diagnostic of it.

Second, the effective parameter count is large, at 292.73 against a nominal 12 for **basic**. This gap measures how much of the apparent sharpness of the composite likelihood arises from treating linked sites as independent (Coffman et al., 2016). Any comparison of uncertainty between the two approaches must account for it, because a method that models linkage, or that draws on independent simulated replicates, does not make the same error and should not be expected to reproduce the same widths.

Third, two of the three fits ended on bounds. Their estimates are therefore conditional on the search box, in a way that a prior over the same range would express differently, and for **growth** there is no interior optimum to compare against at all. The usable baseline is therefore **basic**: its estimates, its interval widths and its correlation matrix, all at $\varepsilon = 0.01$ and all conditional on the model.

4.7 Limitations and extensions

The intervals reported here are conditional on the model, are converted to physical units at a point estimate of N_{ref} whose own interval spans a factor of five, and rest on inverting an H that is numerically singular for **basic**. Part of the width on the deep parameters may therefore be numerical rather than statistical.

Three parameters ended at or near a bound: N_S in **growth** at 49.97 against a bound of 50, m_{12} in **sec_contact** at 50, and $m_{2,ds}$ in **basic** at 18.76 against a bound of 20, with its interval reaching 19.51. Whether wider bounds would help depends on which of the three is considered. For N_S they would not: its interval spans a factor of 132, so the likelihood is nearly flat in that direction and the optimiser would simply run to whatever bound replaced the current one,

returning a different number rather than a better fit. The appropriate diagnostic there is a profile likelihood over N_S (Raue et al., 2009), not a wider box. For the two migration parameters the bound is genuinely constraining the fit, but both already sit at or near the value that the dadi documentation gives as an upper bound for migration rates, so raising the bound would move the fit into a region where evaluation is slow and the accuracy of the diffusion approximation has not been characterised. A refit with wider migration bounds therefore needs a grid-refinement check alongside it, or a boundary problem is simply exchanged for a numerical one. In either case the bound is a property of the search rather than of the data, and a fit that ends on one should be reported as such rather than read as an estimate.

Two chromosomes is also a small amount of sequence. Terhorst and Song (2015) show that the error achievable in demographic inference from a site frequency spectrum decreases no faster than logarithmically in the number of sampled haplotypes, whereas it decreases polynomially in the number of independent loci, so the binding constraint is the amount of sequence rather than the number of animals. Taken together with Myers et al. (2008), who show that distinct size histories can produce identical spectra, this is the formal version of the point made about T_A : the deep parameters are weakly identified in principle, not merely uncertain in practice. Two model assumptions are also violated. No genic masking was applied, and the wild-caught Scottish cats are not a clean, randomly mating group (Howard-McCombe et al., 2023). Both violations would be absorbed by the recent-epoch parameters (Chikhi et al., 2010; Mazet et al., 2016), which are the ones with the narrowest intervals, and neither is reflected in those intervals, since the Godambe matrix is computed under the assumption that the model is correct.

The most direct extension would be to refit the IM model on the spectrum of Section 2.3 under the same schedule and compare it on CLAIC. `sec_contact` imposes a period of isolation that IM does not, so the 806-unit gap measures in part how poorly that assumption fits, rather than how much the historical labels of `basic` add over a generic split-and-migrate model. The second question is the one Aim 1 was originally posed to answer, and it remains open until IM is included in the comparison.

Adding sequence would help more than adding animals. Polarising the spectrum would be the most useful change to the data itself, since folding discards part of the asymmetry between the two populations that the most poorly determined parameters describe. A model with a pulse of admixture would test the hypothesis of Section 4.4 directly, and a profile likelihood over N_S would establish whether `growth` can be compared with `basic` at all.

5 Conclusion

The model with a period of isolation followed by secondary contact scores 806 CLAIC units worse than `basic`. This answers Aim 1, but only in the sense that the data favour more demographic structure than that model allows; it does not show that the historical labels attached to `basic` are correct. Of the two labelled times, T_D falls near the event it is named for, but with an interval too wide to be treated as a date, whereas T_B is placed at a time when nothing relevant is known to have occurred. Its within-model interval is narrow, at 648 to 700 years, and even naively propagating the uncertainty in N_{ref} widens it only to roughly 300 to 1,500 years, a range that contains none of the candidate events. The land-bridge label should be dropped.

Aim 2 is answered by the interval widths and by the correlations behind them. Four parameters are determined to within a factor of two: the recent wildcat size and the time of its change, and

the two recent migration rates. Everything else, including the recent domestic size, is determined only to within a factor of between four and eleven. These widths are not independent: the older parameters lie along a single ridge in the likelihood, and two of the four narrow parameters, T_B and $m_{2,ds}$, are fixed only in combination. The agreement between **basic** and **growth** should not be mistaken for precision, since the two models agree closely on N_{ref} and T_A without either quantity being well determined, and they place T_D a factor of four apart. The **growth** fit did not reach an interior optimum, so it serves as a check on **basic** rather than as a competing estimate.

Aim 3 is met in that a confidence interval is reported for every parameter of all three fits, although two qualifications attach to them. They are within-model intervals: two models that the data cannot separate return intervals of substantially different width for the same quantities, and place T_D a factor of four apart, so model uncertainty appears nowhere in the tables. They are also marginal intervals, whereas what the spectrum actually constrains are combinations of parameters. A simulation-based method checked against the interval widths alone could therefore appear to reproduce this baseline while disagreeing about which combinations the data constrain. The correlation matrix therefore belongs alongside the interval table as part of the baseline. The uncertainty against which simulation-based inference should be checked is both wider and more structured than any single table in this report shows.

References

- Beaumont, M., Barratt, E. M., Gottelli, D., Kitchener, A. C., Daniels, M. J., Pritchard, J. K., and Bruford, M. W. (2001). Genetic diversity and introgression in the Scottish wildcat. *Molecular Ecology*, 10(2):319–336.
- Chernoff, H. (1954). On the distribution of the likelihood ratio. *The Annals of Mathematical Statistics*, 25(3):573–578.
- Chikhi, L., Sousa, V. C., Luisi, P., Goossens, B., and Beaumont, M. A. (2010). The confounding effects of population structure, genetic diversity and the sampling scheme on the detection and quantification of population size changes. *Genetics*, 186(3):983–995.
- Coffman, A. J., Hsieh, P.-H., Gravel, S., and Gutenkunst, R. N. (2016). Computationally efficient composite likelihood statistics for demographic inference. *Molecular Biology and Evolution*, 33(2):591–593.
- Cranmer, K., Brehmer, J., and Louppe, G. (2020). The frontier of simulation-based inference. *Proceedings of the National Academy of Sciences*, 117(48):30055–30062.
- Crow, J. F. and Kimura, M. (1970). *An Introduction to Population Genetics Theory*. Harper & Row, New York.
- Driscoll, C. A., Menotti-Raymond, M., Roca, A. L., Hupe, K., Johnson, W. E., Geffen, E., Harley, E. H., Delibes, M., Pontier, D., Kitchener, A. C., Yamaguchi, N., O’Brien, S. J., and Macdonald, D. W. (2007). The Near Eastern origin of cat domestication. *Science*, 317(5837):519–523.
- Godambe, V. P. (1960). An optimum property of regular maximum likelihood estimation. *The Annals of Mathematical Statistics*, 31(4):1208–1211.
- Gutenkunst, R. N., Hernandez, R. D., Williamson, S. H., and Bustamante, C. D. (2009). Inferring the joint demographic history of multiple populations from multidimensional SNP frequency data. *PLoS Genetics*, 5(10):e1000695.
- Gutenkunst, R. N., Waterfall, J. J., Casey, F. P., Brown, K. S., Myers, C. R., and Sethna, J. P. (2007). Universally sloppy parameter sensitivities in systems biology models. *PLoS Computational Biology*, 3(10):e189.
- Heagerty, P. J. and Lumley, T. (2000). Window subsampling of estimating functions with application to regression models. *Journal of the American Statistical Association*, 95(449):197–211.
- Higham, N. J. (2002). *Accuracy and Stability of Numerical Algorithms*. Society for Industrial and Applied Mathematics, Philadelphia, PA, 2nd edition.
- Howard-McCombe, J., Jamieson, A., Carmagnini, A., Russo, I.-R. M., Ghazali, M., Campbell, R., Driscoll, C., Murphy, W. J., Nowak, C., O’Connor, T., Tomsett, L., Lyons, L. A., Muñoz-Fuentes, V., Bruford, M. W., Kitchener, A. C., Larson, G., Frantz, L., Senn, H., Lawson, D. J., and Beaumont, M. A. (2023). Genetic swamping of the critically endangered Scottish wildcat was recent and accelerated by disease. *Current Biology*, 33(21):4761–4769.e5.
- Howard-McCombe, J., Ward, D., Kitchener, A. C., Lawson, D., Senn, H. V., and Beaumont, M. A. (2021). On the use of genome-wide data to model and date the time of anthropogenic hybridisation: An example from the Scottish wildcat. *Molecular Ecology*, 30(15):3688–3702.

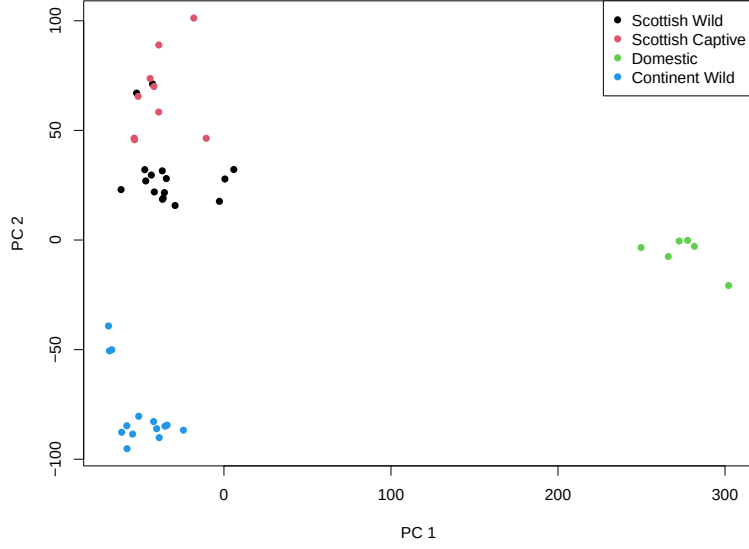
- Johnson, W. E., Eizirik, E., Pecon-Slattery, J., Murphy, W. J., Antunes, A., Teeling, E., and O'Brien, S. J. (2006). The late Miocene radiation of modern Felidae: a genetic assessment. *Science*, 311(5757):73–77.
- Langley, P. J. W. and Yalden, D. W. (1977). The decline of the rarer carnivores in Great Britain during the nineteenth century. *Mammal Review*, 7(3-4):95–116.
- Mazet, O., Rodríguez, W., Grusea, S., Boitard, S., and Chikhi, L. (2016). On the importance of being structured: instantaneous coalescence rates and human evolution—lessons for ancestral population size inference? *Heredity*, 116(4):362–371.
- Myers, S., Fefferman, C., and Patterson, N. (2008). Can one learn history from the allelic spectrum? *Theoretical Population Biology*, 73(3):342–348.
- Nussberger, B., Currat, M., Quilodran, C. S., Ponta, N., and Keller, L. F. (2018). Range expansion as an explanation for introgression in European wildcats. *Biological Conservation*, 218:49–56.
- Oehlert, G. W. (1992). A note on the delta method. *The American Statistician*, 46(1):27–29.
- Portik, D. M., Leaché, A. D., Rivera, D., Barej, M. F., Burger, M., Hirschfeld, M., Rödel, M.-O., Blackburn, D. C., and Fujita, M. K. (2017). Evaluating mechanisms of diversification in a Guineo-Congolian tropical forest frog using demographic model selection. *Molecular Ecology*, 26(19):5245–5263.
- Raue, A., Kreutz, C., Maiwald, T., Bachmann, J., Schilling, M., Klingmüller, U., and Timmer, J. (2009). Structural and practical identifiability analysis of partially observed dynamical models by exploiting the profile likelihood. *Bioinformatics*, 25(15):1923–1929.
- Self, S. G. and Liang, K.-Y. (1987). Asymptotic properties of maximum likelihood estimators and likelihood ratio tests under nonstandard conditions. *Journal of the American Statistical Association*, 82(398):605–610.
- Städler, T., Haubold, B., Merino, C., Stephan, W., and Pfaffelhuber, P. (2009). The impact of sampling schemes on the site frequency spectrum in nonequilibrium subdivided populations. *Genetics*, 182(1):205–216.
- Terhorst, J. and Song, Y. S. (2015). Fundamental limits on the accuracy of demographic inference based on the sample frequency spectrum. *Proceedings of the National Academy of Sciences*, 112(25):7677–7682.
- Transtrum, M. K., Machta, B. B., Brown, K. S., Daniels, B. C., Myers, C. R., and Sethna, J. P. (2015). Perspective: Slowness and emergent theories in physics, biology, and beyond. *The Journal of Chemical Physics*, 143(1):010901.
- Varin, C., Reid, N., and Firth, D. (2011). An overview of composite likelihood methods. *Statistica Sinica*, 21(1):5–42.
- Varin, C. and Vidoni, P. (2005). A note on composite likelihood inference and model selection. *Biometrika*, 92(3):519–528.
- Vigne, J.-D., Briois, F., Zazzo, A., Willcox, G., Cucchi, T., Thiébaud, S., Carrère, I., Franel, Y., Touquet, R., Martin, C., Moreau, C., Comby, C., and Guilaine, J. (2012). First wave of cultivators spread to Cyprus at least 10,600 y ago. *Proceedings of the National Academy of Sciences*, 109(22):8445–8449.

- Wang, R. J., Raveendran, M., Harris, R. A., Murphy, W. J., Lyons, L. A., Rogers, J., and Hahn, M. W. (2022). De novo mutations in domestic cat are consistent with an effect of reproductive longevity on both the rate and spectrum of mutations. *Molecular Biology and Evolution*, 39(7):msac147.
- Weninger, B., Schulting, R., Bradtmöller, M., Clare, L., Collard, M., Edinborough, K., Hilpert, J., Jöris, O., Niekus, M., Rohling, E. J., and Wagner, B. (2008). The catastrophic final flooding of Doggerland by the Storegga Slide tsunami. *Documenta Praehistorica*, 35:1–24.
- Wright, S. (1938). Size of population and breeding structure in relation to evolution. *Science*, 87(2263):430–431.

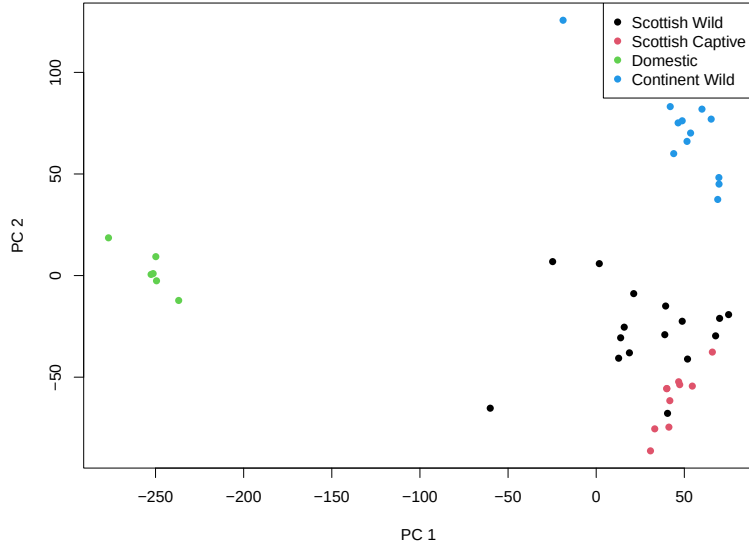
A Code availability

All code used in this report is available at https://github.com/timj-wood/wildcat_models.

B PCA plots



(a) A1 Chromosome (NC_018723.3).



(b) A2 Chromosome (NC_018724.3).

Figure A1: Principal components 1 and 2 for all 46 individuals, computed separately for each chromosome from the SNP data described in Section 2.1. Points are coloured by the four groups of Table 1. In both panels the first component separates the domestic cats from all wildcats, and the second separates the mainland European from the Scottish animals, which matches the order of differentiation given by the pairwise F_{ST} values in Section 2.2. The wild-caught and captive Scottish animals are partly separated on the second component in (a) and overlap in (b). Component signs are arbitrary and differ between panels, so the same group appears on opposite sides in (a) and (b).

C Convergence diagnostics

Table A1: Round-by-round optimisation for the three models. The gain is the improvement in log-likelihood over the previous round, and the spread is the gap between the best and tenth-best restart within the round. Every round used 50 restarts, all of which returned a fit. The polished row is the re-optimisation of the round-4 point that was carried out before the Godambe matrices were computed.

Model	Round	ℓ_{CL}	Gain	Top-10 spread
basic	1	-1416.7730	—	125.834
	2	-1411.2542	+5.519	0.545
	3	-1411.1749	+0.079	0.241
	4	-1411.1464	+0.029	0.107
	polished	-1411.14	+0.006	—
growth	1	-1426.9120	—	80.250
	2	-1396.6429	+30.269	5.505
	3	-1383.2923	+13.351	4.855
	4	-1376.7944	+6.498	0.903
	polished	-1376.22	+0.574	—
sec_contact	1	-1768.6865	—	0.315
	2	-1768.6863	+0.0003	0.00031
	3	-1768.6863	-0.0000	0.00023
	4	-1768.6863	+0.0000	0.00025
	polished	-1768.69	—	—

Table A2: Values of the two **growth** parameters that did not settle, across optimisation rounds. N_S has an upper bound of 50.

Parameter	Round 1	Round 2	Round 3	Round 4	Polished
N_S	0.83	4.62	22.14	47.07	49.97
T_A	5.93	8.29	7.76	6.21	6.08

D Correlation matrices for growth and sec_contact

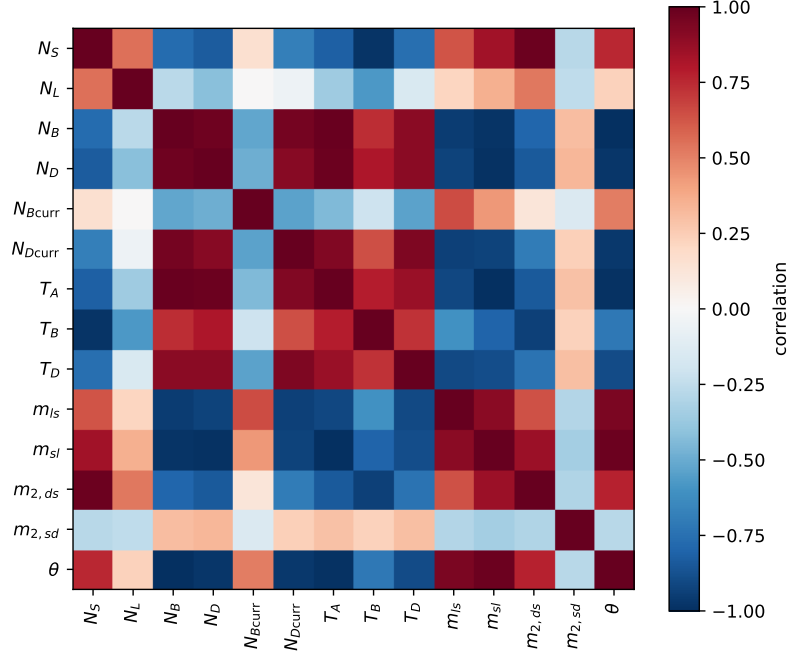


Figure A2: Correlation matrix for **growth** at $\varepsilon = 0.01$, constructed as in Figure 6. Parameter order follows the model function, so N_{Bcurr} and N_{Dcurr} appear fifth and sixth rather than last as in Table 6. This fit terminated on a bound, so the matrix describes the curvature of the likelihood at that point rather than at a maximum.

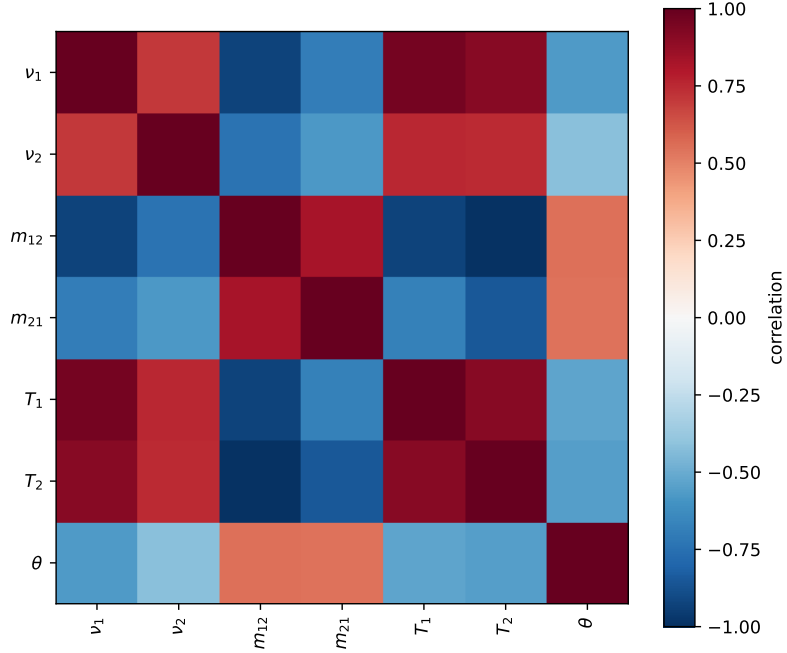


Figure A3: Correlation matrix for **sec_contact** at $\varepsilon = 0.01$, constructed as in Figure 6. m_{12} finished on its upper bound, so its row and column are subject to the same caveat as Figure A2.